

Flappy Bird - Interim Short Report

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1. Game Strategy

The game is set up to have two modes: 'TRAINING' and 'GAME' which have different strategies.

1.1 General Strategy

For both modes, the fundamental concept of the game remains in place of a bird needing to successfully navigate their way through pipes in order to gain points. The bird will be in free fall until a mouse click is detected, in which the bird will fly up by a certain vertical distance.

The game ends in both modes when the bird has run out of lives. Lives can be lost either by the bird hitting a pipe, or hitting the ground. Hitting a pipe will result in the loss of one life, while hitting the ground will result in the loss of all remaining lives.

1.2 TRAINING Mode

In training mode, the difficulty of the game is set at its lowest level and will remain at this level the entire time.

In this mode, the bird will have three lives and the game will end once all lives are lost.

The current speed of the pipes will have them moving one pixel to the right for every vertical synchronisation cycle.

Unlike in the GAME mode, there will be no additional items to collect.

1.3 GAME Mode

The main difference in this mode compared to TRAINING is that the bird will only have one life the entire time.

In the game mode, there will be three distinct levels of difficulty for the game. The difficulty level will be determined by the score, with the level increasing every 20 until the third level. The increase in difficulty is implemented through the speed at which the pipes are moving, which increase with each level.

In this mode there is a possibility for objects to spawn within the middle of the pipes which the bird can collect. The object spawned will either be a coin object. Collecting the coin object will add an additional point to the score.

2. Design Specifications

2.1 Text

Text on the VGA display is first seen in the start screen of the game, when it is initially loaded. The text on the start screen is white, which is the same for text on all screens.

For both modes, there is the name of the current mode near the top with a 'Score' text underneath.

The text on screen on the VGA display is implemented through the char_rom component of the project

2.2 Time

Time was decided to be displayed on the seven-segment LED display rather than on the VGA screen. This decision stemmed from trying to keep the screen more minimalist to avoid distractions to the player.

Time is implemented through an upCounter component and supplies output to the HEX2 to HEX0 LED displays.

2.3 Changing Screens

To change screens the user must either switch down SW9 for training mode or SW8 for game mode and then press Button0 on the DE0 board to go into either game screen. The start function of the pressed Button0 is used in both char_rom and objects components.

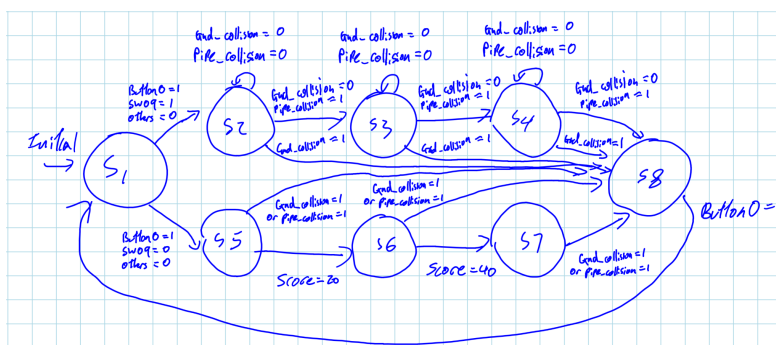
2.4 Objects

There are currently two objects which have been implemented into the project, those being the bird and pipe.

The bird can only be seen once the user has selected a game mode and started the game. It is currently displayed as a magenta square on the left side of the screen.

The pipe can be seen from the start screen as an object moving from right to left of the screen. After a game mode has started there will be a singular gap in the pipe for each iteration of the pipe. The pipe is displayed as two green rectangles.

Both objects are in the objects component as they will need to interact with each other when collisions between the two are implemented.



2.5 Mouse

The purpose of the mouse is to serve as an interface between the user and the game itself. Upon clicking the mouse button, this would send a signal to the object component which

would cause the bird to increase by a specific height across a period of clock cycles.

3.0 Plans

3.1 Game Mechanics

Collisions and lives will need to be implemented into the game for both modes. These will likely be implemented in the objects component as it contains both the bird and pipe objects and their movements.

3.2 Summary Screen

We plan to implement a summary screen with the text of 'Summary' and the score that the player got. Underneath both the summary and score text will be 'Title Screen' with the instruction on how to return to the title screen.

